Phase 2 — ER Modeling

Campus Complaint & Grievance Tracker — Entity-Relationship Design & Table Design

Entities & Attributes

Each 'thing' the system needs to remember becomes an entity. We identified nine entities:

Entity	Purpose
User (superclass)	Common identity data shared by every person in the system
Student (subclass)	Files complaints; belongs to a hostel and room
Staff (subclass)	Resolves complaints; also covers Warden and Admin via role_type
Hostel	Groups students; has one warden
Department	Owns one or more categories; employs staff
Category	Hostel/Mess/Academics/Infra/Lab/Library, each with its own SLA
Complaint	The core record — one row per grievance filed
StatusHistory	Audit trail of every status change on a complaint
Staff_Assignment	Which staff member is/was handling which complaint

Generalization / Specialization (User → Student, Staff)

Rather than creating one flat table with columns for every role (which would leave 'roll_number' empty for staff and 'designation' empty for students), User is modeled as a superclass holding only what every person shares (name, email, phone). Student and Staff are subclasses holding just their role-specific attributes, linked back to User by a shared primary key (user_id).

- **Disjoint:** a user is exactly one of Student or Staff — never both.
- **Total:** every user row must have a matching Student or Staff row — there is no 'unclassified' user.
- **Why Staff covers Warden and Admin too:** Warden and Admin are still campus employees, not students, so they belong under Staff. A role_type column (staff / warden / admin) distinguishes them without needing a third subclass table — this keeps the schema simple while still letting your application logic (and views in Phase 6) treat them differently.

Relationships & Cardinality

Relationship	Cardinality	Meaning
Hostel → Student	1 : M	One hostel houses many students
Staff → Hostel	1 : 0..1	A staff member (role_type=warden) manages at most one hostel
Department → Staff	1 : M	One department employs many staff
Department → Category	1 : M	One department owns several complaint categories
Category → Complaint	1 : M	One category classifies many complaints
Student → Complaint	1 : M	One student can file many complaints
Complaint → StatusHistory	1 : M	Every status change is logged as a new row

Relationship	Cardinality	Meaning
User → StatusHistory	1 : M	Records who made each change (nullable for system/auto-escalation)
Complaint → Staff_Assignment	1 : M	Full reassignment history is kept, not overwritten
Staff → Staff_Assignment	1 : M	One staff member can be linked to many assignments over time

Note: Complaint intentionally does NOT store a hostel_id directly. A complaint's hostel can always be derived by joining Complaint → Student → Hostel. Storing it a second time on Complaint would be a transitive dependency and a normalization violation

The ER Diagram

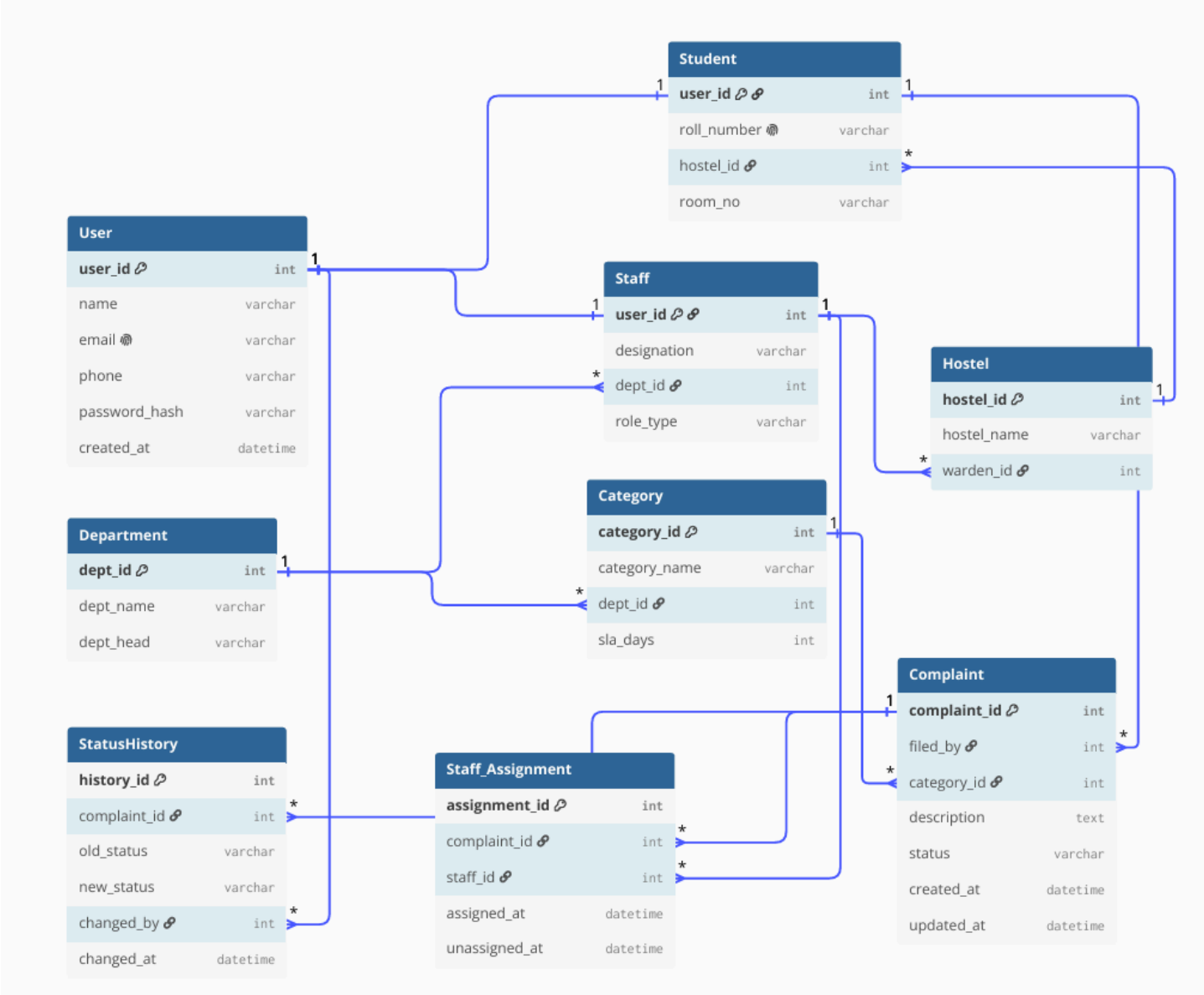


Table Design (Relational Schema)

User (superclass)

Column	Data Type	Constraint
user_id	INT	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	UNIQUE, NOT NULL
phone	VARCHAR(15)	
password_hash	VARCHAR(255)	NOT NULL
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

Student (subclass of User)

Column	Data Type	Constraint
user_id	INT	PRIMARY KEY, FOREIGN KEY -> User(user_id)
roll_number	VARCHAR(20)	UNIQUE, NOT NULL
hostel_id	INT	FOREIGN KEY -> Hostel(hostel_id)
room_no	VARCHAR(10)	

Staff (subclass of User)

Column	Data Type	Constraint
user_id	INT	PRIMARY KEY, FOREIGN KEY -> User(user_id)
designation	VARCHAR(50)	
dept_id	INT	FOREIGN KEY -> Department(dept_id)
role_type	ENUM('staff','warden','admin')	NOT NULL, DEFAULT 'staff'

Hostel

Column	Data Type	Constraint
hostel_id	INT	PRIMARY KEY, AUTO_INCREMENT
hostel_name	VARCHAR(50)	NOT NULL
warden_id	INT	FOREIGN KEY -> Staff(user_id), NULLABLE

Department

Column	Data Type	Constraint
dept_id	INT	PRIMARY KEY, AUTO_INCREMENT
dept_name	VARCHAR(50)	NOT NULL
dept_head	VARCHAR(100)	

Category

Column	Data Type	Constraint
category_id	INT	PRIMARY KEY, AUTO_INCREMENT
category_name	VARCHAR(50)	NOT NULL
dept_id	INT	FOREIGN KEY -> Department(dept_id)
sla_days	INT	NOT NULL, DEFAULT 3

Complaint

Column	Data Type	Constraint
complaint_id	INT	PRIMARY KEY, AUTO_INCREMENT
filed_by	INT	FOREIGN KEY -> Student(user_id)
category_id	INT	FOREIGN KEY -> Category(category_id)
description	TEXT	NOT NULL
status	ENUM('Open','Assigned','In-Progress','Resolved','Escalated')	DEFAULT 'Open'
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP
updated_at	DATETIME	ON UPDATE CURRENT_TIMESTAMP

StatusHistory

Column	Data Type	Constraint
history_id	INT	PRIMARY KEY, AUTO_INCREMENT
complaint_id	INT	FOREIGN KEY -> Complaint(complaint_id)
old_status	VARCHAR(20)	
new_status	VARCHAR(20)	
changed_by	INT	FOREIGN KEY -> User(user_id), NULLABLE
changed_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

Staff_Assignment

Column	Data Type	Constraint
assignment_id	INT	PRIMARY KEY, AUTO_INCREMENT
complaint_id	INT	FOREIGN KEY -> Complaint(complaint_id)
staff_id	INT	FOREIGN KEY -> Staff(user_id)
assigned_at	DATETIME	DEFAULT CURRENT_TIMESTAMP
unassigned_at	DATETIME	NULLABLE